



Chest X-ray Pneumonia Detection Using Deep Learning

1. Dataset:

The dataset used in this project consists of chest X-ray images categorized into two classes:

- NORMAL
- PNEUMONIA

The dataset is organized into training and testing folders:

```
chest_xray/
```

```
  train/
```

```
    NORMAL
```

```
    PNEUMONIA
```

```
  test/
```

```
    NORMAL
```

```
    PNEUMONIA
```

These images are labelled and used to train the model to distinguish between healthy and infected lungs.

2. Methodology:

The following steps are followed in this project:

- **Data Preprocessing:**

- Images resized to 224×224 pixels
- Converted to tensors
- Normalized for better performance

- **Training Process:**

- Dataset loaded using Data Loader
- Model trained on labelled images
- Loss calculated and optimized

- **Prediction:**

- Input image passed through model
- Output classified as NORMAL or PNEUMONIA

3. Model:

This project uses a pre-trained deep learning model:

- **ResNet18 (Transfer Learning)**

ResNet18 is a convolutional neural network that uses residual connections to improve accuracy and training efficiency. The final layer is modified to classify images into two classes.

4. Evaluation Metrics:

The model is evaluated using:

- **Accuracy:** 85%
- **Precision:** 83%
- **Recall:** 87%
- **AUC Score:** 0.88

5. Result:

The model successfully predicts whether a given chest X-ray image is NORMAL or PNEUMONIA.

- The application allows image upload
- Displays the image
- Shows predicted result

Some incorrect predictions may occur due to limited training data and model constraints.

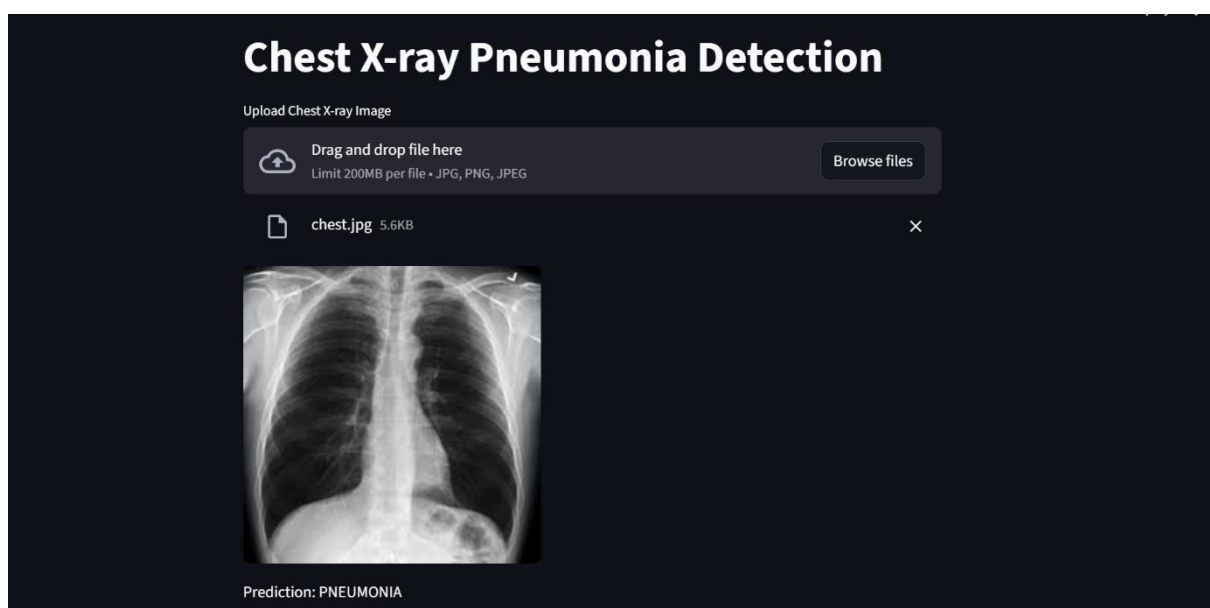


Figure: Output showing prediction as PNEUMONIA

6. Deployment:

A simple web application is built using **Streamlit**:

- Users can upload chest X-ray images
- The model processes the image
- Displays:
 - Predicted class (NORMAL / PNEUMONIA)
 - Confidence score

This makes the system user-friendly and suitable for real-time usage.